

# Henrique Ennes

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## Curriculum Vitae

### Personal information

Full name Henrique Lovisi Ennes  
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GitHub [HLovisiEnnes](https://github.com/HLovisiEnnes)  
Website <https://hlovisiennes.github.io/>

### Education

- 2023 – 2026 **Ph.D. candidate in Computer Science**, *Université Côte d'Azur/Inria*, Sophia Antipolis, France.  
Advisors: Clément Maria and Nicolas Nisse.  
Thesis title: *Quantum computing with Heegaard splittings of 3-manifolds*.  
Defense scheduled for 29/09/2026.
- 2021 – 2023 **Master's degree in Applied Mathematics**, *Fundação Getulio Vargas - EMAP*, Rio de Janeiro, Brazil.  
Advisors: Raphaël Tinarrage and César Camacho.  
GPA: 4.00/4.00.
- 2016 – 2020 **Bachelor of Arts, Mathematics and Physics (*summa cum laude*)**, *Whitman College*, Walla Walla, USA.  
Honors in majors.  
GPA: 3.97/4.00.

### Journal publications

- 2026 **Hardness of computation of quantum invariants on 3-manifolds with restricted topology**, with Clément Maria.  
*Journal of Applied and Computational Topology*.
- 2025 **LieDetect: Detection of representation orbits of compact Lie groups from point clouds**, with Raphaël Tinarrage.  
*Foundations of Computational Mathematics*.
- 2025 **Empirical analysis of Binding Precedent efficiency in the Brazilian Supreme Court via Similar Case Retrieval**, with Raphaël Tinarrage, Lucas E. Resck, Lucas T. Gomes, Jean R. Ponciano, and Jorge Poco.  
*Artificial Intelligence and Law*.
- 2021 **Two-body bound states through Yukawa forces and perspectives on hydrogen and deuterium**, with Moira Gresham and Alexander Shaw.  
*American Journal of Physics*.

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## Peer-reviewed conference papers

- 2026 **Compressed data structures for Heegaard splittings**, with Clément Maria.  
*42nd International Symposium on Computational Geometry (SoCG 2026)*.  
Invited to submit an extended version to a special issue of *Discrete & Computational Geometry*.
- 2025 **Hardness of computation of quantum invariants on 3-manifolds with restricted topology**, with Clément Maria.  
*33rd European Symposium of Algorithms*.
- 2024 **Cost Benefit Analysis for Investments in Power Grid Resilience - A Guide**, with Edson Daniel Lopes Gonçalves, Joisa Dutra, Rafael Souza.  
*9th Latin American Energy Economics Meeting (ELAEE)*.
- 2023 **Measuring the Power Grid Resilience: A Case Study Applied to Brazilian Distribution Companies**, with Joisa Dutra, Rafael Souza, Rafael Gomes, Lucas Amaro, Camila Albertin.  
*27th International Conference on Electricity Distribution*.

## Preprints

- 2026 **Theory of Trotter Error with Variable-Weight Schedules**.  
Preprint.
- 2026 **Removing Online Exponential Net Search from Solovay-Kitaev**, with Clément Maria.  
Preprint.
- 2025 **Variational Quantum Brushes**, with Jui-Ting Lu, Chih-Kang Huang, and Ali Abbassi.  
Preprint.

## Selected presentations

- Jun 2026 **Compressed data structures for Heegaard Splittings**, *42nd SoCG 2026*, New Brunswick (NJ), USA.
- Sep 2025 **Hardness of computation of quantum invariants on 3-manifolds with restricted topology**, *33rd ESA*, Warsaw, Poland.
- Aug 2025 **Quantum computing (with knots)**, *CEMRACS*, Marseille (CIRM), France.
- May 2024 **From fields to topology: constructing TQFT invariants through Physics**, *2024 DataShape Workshop*, Porquerolles, France.
- Jan 2024 **Detection of representation orbits of compact Lie groups from point clouds**, *DataShape Seminar*, Sophia Antipolis, France.
- Jul 2023 **Detection of compact Lie group representations in point clouds and image data**, *2023 TDA week*, Kyoto, Japan.
- Mar 2023 **Debater at the Workshop "Transforming the Role of International Courts and Tribunals in a New Era of Adjudication"**, *ERASMUS+ Jean Monnet Centre of Excellence*, Rio de Janeiro, Brazil.

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Nov 2018 **Simulation of bound states of dark matter through Yukawa potentials**, *Murdock Conference*, Vancouver (WA), USA.

## Teaching and mentoring experience

Feb 2025 **Lecturer: Introduction to Data Science and Machine Learning**, *Fundação Getúlio Vargas, Brazilian Ministry of Mines and Energy and World Bank (Project META)*, Rio de Janeiro, Brazil (online).  
Apr 2025  
Sep 2022 **Lecturer: Introduction to Mathematical Modelling Applied to Law**, *Fundação Getúlio Vargas – ERASMUS+ Jean Monnet Centre of Excellence*, Rio de Janeiro, Brazil.  
Dec 2022  
Aug 2020 **Teaching Assistant: Physics 347 (Classical Mechanics)**, *Whitman College*, Walla Walla, USA (remote).  
Dec 2020  
Jan 2018 **Tutor: Mathematics 125, 126, and 225 (Calculus)**, *Whitman College*, Walla Walla, USA.  
May 2020  
Aug 2017 **Tutor: Physics 155 and 156 (Introductory Physics Courses)**, *Whitman College*, Walla Walla, USA.  
May 2020

## Academic service and outreach

2025 Developer for a **digital-art project** in quantum-computing outreach.  
2023 Reviewer for **Symposium on Computational Geometry (SoCG)**.  
2018 – 2020 Physics and mathematics departments' **student representative**.  
2017 Founder of **STEM4ALL**, an outreach initiative in the quantitative sciences.

## Grants and awards

PhD **QuanTEdu France grant**  
Offered by the French government to foster the development of quantum technologies.  
Master's **CAPES graduate student grant**  
Offered by the Ministry of Education based on merit criteria.  
Undergraduate **Elected member of Phi Beta Kappa honor society**  
Election based on academic merit.  
**Laura and John Hook Family Mathematics Award**  
Awarded to a graduating senior in Mathematics for demonstrated talent.  
**Student Commencement Marshal**  
Chosen among the 10 highest general GPAs of the junior class (GPA: 3.97/4.00).  
**M&G Wier Scholarship**  
Awarded to a junior student in recognition of their talent in Mathematics.  
**Tristram S. Lundquist Scholarship Endowment**  
Awarded to a Physics major student in recognition of academic performance.  
Others **Mayor's medal of honor for advancing the study of Astronomy in Juiz de Fora**  
For high performance in the Brazilian Astronomy Olympics.

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## References

PhD Clément Maria (Inria d'Université Côte d'Azur, France)  
supervisor clement.maria@inria.fr

PhD Nicolas Nisse (Inria d'Université Côte d'Azur, France)  
supervisor nicolas.nisse@inria.fr

Master's Raphaël Tinarrage (ISTA, Austria)  
supervisor raphael.tinarrage@ist.ac.at

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